the extra Calabi-Yau dimensions determines whether certain physical configurations within string theory appear as black holes or elementary particles. That is, in the first phase, the initial Calabi-Yau shape (the analog of the ice phase, say), we find that there are certain black holes present. In the second phase, the second Calabi-Yau shape (the analog of the liquid water phase), these black holes have gone through a phase transition-they have “melted” so to speak-into fundamental vibrational string patterns. The tearing of space through conifold transitions takes us from one Calabi-Yau phase to the other. In so doing, we see that black holes and elementary particles, like water and ice, are two sides of the same coin. We see that black holes snugly fit within the framework of string theory.

We have purposely used the same water analogy for these drastic space-tearing transmutations and for the transmutations from one of the five formulations of string theory to another (Chapter 12) because they are deeply connected. Recall that we expressed through Figure 12.11 that the five string theories are dual to one another and thereby are unified under the rubric of a single overarching theory. But does the ability to move continuously from one description to another—to set sail from any point on the map of Figure 12.11 and reach any other —persist even after we allow the extra dimensions to curl up into some Calabi-Yau shape or another? Prior to the discovery of the drastic topology-changing results, the anticipated answer was no, since there was no known way to continuously deform one Calabi-Yau shape into any other. But now we see that the answer is yes: Through these physically sensible space-tearing conifold transitions, we can continuously change any given Calabi-Yau space into any other. By varying coupling constants and curled-up Calabi-Yau geometry, we see that all string constructions are, once again, different phases of a single theory. Even after curling up all extra dimensions, the unity of Figure 12.11 firmly holds.

Black Hole Entropy

For many years, some of the most accomplished theoretical physicists speculated about the possibility of space-tearing processes and of a connection between black holes and elementary particles. Although such speculations might have sounded like science fiction at first, the discovery of string theory, with its ability to merge general